

ProbMap: Automatically constructing design galleries through feature extraction and semantic clustering

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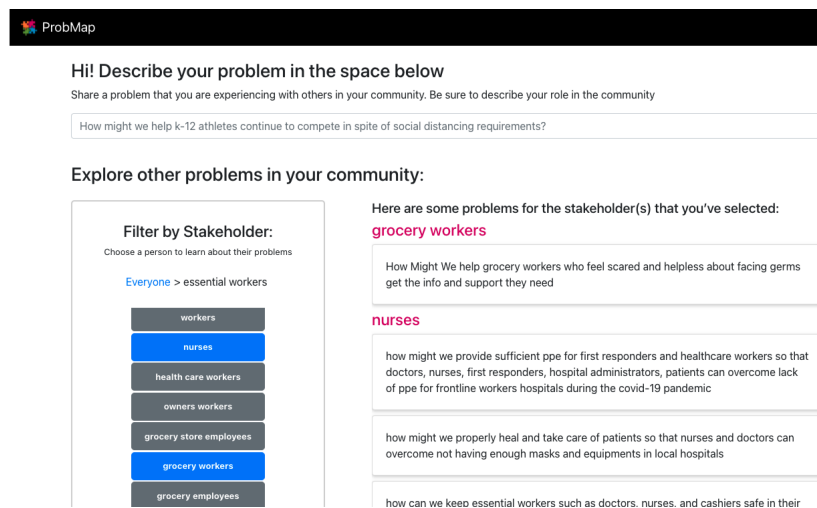


Figure 1: ProbMap is a design gallery for problem statements. Stakeholders are extracted from problem statements and then used as search facets to guide exploration. This figure shows problem statements for the stakeholder cluster ‘essential workers’.

ABSTRACT

Making sense of large unstructured problem spaces is cognitively demanding. Structure can help, but adding structure to a problem space also takes significant effort. ProbMap is a novel application for automatically constructing a design gallery from unstructured text input. Given a list of problem statements, ProbMap extracts and semantically groups the stakeholders to construct hierarchical search facets which enables designers to more efficiently navigate the problem statements. We contribute a novel feature extraction algorithm using natural language processing and a technique for automatically constructing a design gallery. These stakeholders are grouped semantically by clustering stakeholders with higher

pairwise similarity together. Preliminary trials show that these techniques, which mirror traditional design activities like stakeholder identification and affinity mapping, provide an initial structure to a large unstructured problem space. This resulted in similar features that would be extracted by humans and sensible clusters.

CCS CONCEPTS

• **Applied computing** → **Document management and text processing**; • **Human-centered computing** → **Collaborative and social computing**.

KEYWORDS

design galleries, feature extraction, design data, search facets

ACM Reference Format:

Stephen MacNeil, Zijian Ding, Kexin Quan, Ziheng Huang, Kenneth Chen, and Steven P. Dow. 2021. ProbMap: Automatically constructing design galleries through feature extraction and semantic clustering. In *The Adjunct Publication of the 34th Annual ACM Symposium on User Interface Software and Technology (UIST '21 Adjunct)*, October 10–14, 2021, Virtual Event, USA. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3474349.3480203>

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UIST '21 Adjunct, October 10–14, 2021, Virtual Event, USA

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ACM ISBN 978-1-4503-8655-5/21/10.

<https://doi.org/10.1145/3474349.3480203>

1 INTRODUCTION

As the world continues to face larger and more complex problems—such as pandemics, income inequality, racism, and climate change—designers need tools and systems to help them get an overview of the problems facing their neighborhood and communities. However, collecting, curating, and making sense of large-scale community input is still a major challenge [8, 14]. Navigating a large complex problem space can be cognitively demanding—even for expert designers [24]. Prior work shows how design galleries can summarize diverse design artifacts such as posters [9], solutions [22], websites [6, 12, 16, 21], visualizations [15], images [5], works of art [10], mobile apps [13], products [6, 18], and generative design concepts [17]. Design galleries provide structure to the design space and designers tend to benefit when ideas are grouped semantically into clusters, rather than being presented without structure [19].

Often, users navigate design galleries through faceted search. For example, D.Tour curates design websites by the amount of text on the page, the primary colors, and the visual composition [21]. Search facets are often derived from the design artifacts themselves through manual annotations [3] or feature extraction [21]. However, feature extraction methods, and design galleries more broadly, have focused on visual design contexts, and less so on text-based features.

We introduce ProbMap a text-based design gallery that is generated automatically by extracting stakeholders and using them as search facets to help designers navigate large complex problem spaces. Our approach leverages natural language processing and clustering to facilitate problem space exploration. By clustering stakeholders into semantic groups, we also provide hierarchical search facets to help designers see the design space at multiple levels of abstraction and reduce the dimensionality of the design space. We make the following contributions: 1) a novel design gallery that focuses on presenting problem statements, 2) a feature extraction method to identify stakeholders from unstructured problem statement, 3) a method to automatically affinity map stakeholders through semantic clustering.

2 SYSTEM OVERVIEW AND CORPUS

ProbMap, shown in Figure 1, is an algorithm and system for curating unstructured text-based design data, such as problem statements. ProbMap automates common design activities completed by expert designers such as stakeholder identification and affinity mapping.

We collected a corpus of 679 problem statements from a series of community design workshops and class assignments related to COVID-19. From these statements, ProbMap extracted 248 distinct stakeholders which were clustered into 68 stakeholder groups. The following sub-sections detail our approach.

2.1 Feature Extraction (Identify Stakeholders)

Stakeholders are the people who affect or are affected by a problem or solution. Identifying stakeholders is an essential starting point for any human-centered design process [4, 20]. Previous research has focused on extracting key design information from products [7], research papers [1, 2], and websites [21]. However, extracting design information about potential users and stakeholders might help to focus design activities toward a human-centered perspective.

To automatically extract stakeholders from our corpus of problem statements, we developed a novel feature extraction method shown in Figure 2. Our approach starts by identifying noun and noun phrases using Part-of-Speech Tagging algorithms provided by the NLTK¹ and TextBlob Python packages². Next, we lemmatized the noun to resolve entities such as *Merchant* and *Merchants*. Finally, we use ConceptNet to determine if each noun is a stakeholder. ConceptNet is a knowledge graph that connects words and phrases [23] through conceptual relationships. For each extracted noun, we recursively traverse ConceptNet through *IsA* relationships to determine if the root concept is *Person*. As an example, (*Merchant*) – [: *isA*] – > (*Person*).

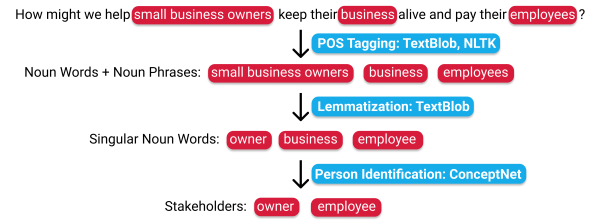


Figure 2: Our feature extraction method for identifying stakeholders within problem statements.

To evaluate, we randomly selected 50 problem statements from our corpus and labeled the stakeholders. There was moderate agreement ($\alpha_u=0.63$) between the labels from a human coder and our feature extraction algorithm as measured by Krippendorff's α_u [11]—a measure of inter-rater reliability for segmentation (i.e.: unitization).

2.2 Clustering (Affinity Mapping)

Designers often engage in sense-making activities, such as affinity mapping, to distill insights from large collections of unstructured or semi-structured user research. These activities focuses the designer's attention on the most important design aspects.

To mimic this affinity mapping process, we developed an algorithm for clustering stakeholders into semantically similar groups. Our algorithm first creates an adjacency matrix based on the pairwise similarity of each stakeholder. Pairwise similarity was computed from the cosine similarity of the word embeddings using the Universal Sentence Encoder³. The resulting adjacency matrix represents an undirected and unweighted graph where edges were drawn between vertices when the similarity was above a threshold of 0.55. Finally, Markov Clustering (MCL) automatically identifies clusters based on a simulation of stochastic flow of the graph [25, 26] using the Markov Clustering Python library⁴. Finally, we select the cluster label by choosing the most common noun phrase.

3 DISCUSSION

Based on our initial evaluation, ProbMap had moderate agreement with human coders related to stakeholder identification. The clusters also appear to provide sensible groupings. For example, one of

¹<https://www.nltk.org/>

²<https://github.com/sloria/textblob>

³<https://tfhub.dev/google/universal-sentence-encoder/1>

⁴<https://markov-clustering.readthedocs.io>

the clusters labeled ‘students’ includes *college students*, *international students*, *freshman students*, *teachers*, and other semantically similar stakeholders. The ‘city leaders’ cluster contained *city leaders*, *politicians*, *government officials*, *mayors*, and others. Overall, the clusters were relatively diverse and included *essential workers*, *students*, *elderly people*, *city leaders*, *business owners*, *homeless*, *patients*, *drivers*, and *parents*. This demonstrate the depth and breadth of stakeholders that ProbMap is capable of curating. This demo paper presents a new method for making sense of unstructured text-based design data, specifically problem statements, by automating design activities that are traditionally completed by expert designers.

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